

PRAKASH TM

Senior Software Engineer | Gen AI

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SUMMARY

Experienced Full Stack Developer (MERN) with 5+ years of experience in designing and delivering scalable, high-performance, SSR & CSR web applications and cloud-native solutions on AWS and Azure. Proven experience in building AI-driven applications using AI agents, RAG, MCP and ChromaDB with strong expertise in end-to-end product development, performance optimization and deploying production-ready systems in Agile environments

- Strong expertise in **JavaScript** and **TypeScript**, with backend development using **Node.js**, **Express.js**, **Nest.js** and **Python (FastAPI)**, and frontend development using **React.js**, **Next.js**, **Streamlit**, **Tailwind CSS**, and **Material UI** to build responsive and scalable user interfaces.

- Experienced in building **RESTful APIs**, **microservices**, and real-time applications using **WebSockets** and **SignalR**, with a focus on performance optimization, scalability, and clean architecture.

- Hands-on experience in **state management** and data-fetching solutions using **Redux Toolkit**, **React Query** and **Cube.js** to improve application performance and maintainability.

- Strong experience with **MongoDB** (aggregation, indexing, Atlas) and working knowledge of **SQL databases** for efficient data modeling and query optimization.

- Practical experience in **AI-driven applications**, including **AI agents**, **Agentic AI**, **Retrieval-Augmented Generation (RAG)**, **Model Context Protocol (MCP)**, and vector databases such as **ChromaDB**.

- Extensive cloud experience with **AWS** (IAM, Lambda, API Gateway, DynamoDB, S3, RDS, EC2, CloudWatch) and **Azure** (Function Apps, Logic Apps, App Services, Storage Accounts, Cosmos DB, API Management, Service Bus, Notification Hub, App Insights, ACI, ACR, SWA).

- Experience in unit and integration testing using **Jest** and **Vitest**, following TDD and best coding practices to ensure code quality and reliability.

- Proficient in **Git**, **GitHub**, **Bitbucket**, **Docker** and **CI/CD pipelines**, working in **Agile/Scrum** environments using **Jira** and **Azure DevOps**.

- Strong **analytical**, **problem-solving**, and **communication skills**, with experience collaborating with cross-functional teams and stakeholders to deliver business-critical solutions.

TECHNICAL SKILLS

- React.js, Next.js
- JavaScript, TypeScript, Python
- Tailwind CSS, Material UI
- Redux Toolkit, Cube.js
- Node.js, Express.js, Next.js, Serverless, Microservices
- FastAPI, REST APIs
- WebSockets, SignalR
- MongoDB, SQL, Postgresql
- AI Agents, RAG, MCP, ChromaDB
- Ollama, LangChain, LangGraph, CrewAI, Hugging Face
- Claude, Copilot, Codex, Cursor
- AWS (IAM, Lambda, API Gateway, DynamoDB, S3, RDS, EC2, CloudWatch)
- Azure (Function & Logic Apps, App Services, Storage Accounts, Cosmos DB, Service Bus, Notification Hub, MSAL, ACI, ACR, SWA)
- CI/CD, Git, GitHub, Bitbucket, Docker
- Jest, Vitest
- JWT
- Jira, Azure DevOps, Zoho,
- Agile/Scrum

EDUCATION

Bachelor of Electronics and Communication Engineering

06/2017 - 04/2021

WORK EXPERIENCE

CESIT PVT LTD, CHENNAI - INDIA

Senior Software Engineer — 05/2026-NOW

Precheck - Web Relay Check In Check Out | Transport

- **Role:** Full Stack Developer | **Tech:** React, Node.js
- **Description:** Developed a logistics management portal, **reducing driver wait time by 35%** and **improving gate processing efficiency by 30%** through automated queue management and Web Relay integration.

Software Engineer — 05/2025-05/2026

DDI Gamification | Operations

- **Role:** Frontend Developer | **Tech:** Next.js, Tailwind CSS
- **Description:** Developed a responsive gamified dashboard to track KPIs, achievements, and billing. **Improved user engagement by 25%** through interactive performance visualization.

MediScanner | Healthcare

- **Role:** Full Stack Developer | **Tech:** React, TypeScript, FastAPI, Python, MongoDB Atlas, Gemini AI, Ollama.
- **Description:** Developed an AI-driven healthcare platform, **reducing manual medical data entry by 60%** using Gemini AI while **improving patient health insight visualization by 35%**.

Arch and monArch | Manufacturing (IoT)

- **Role:** Full Stack Developer | **Tech:** React, Node.js, Moleculer.js, Cube.js, MongoDB, SQL, Azure.
- **Description:** Engineered a real-time factory monitoring platform, **increasing operational efficiency by 40%** and **reducing issue resolution time by 30%** through analytics-driven dashboards and automated workflows.

Software Technologist — 05/2024-05/2025

Metrics | Manufacturing (FMCG)

- **Role:** UI Developer | **Tech:** React, TypeScript, React Flow, SCSS, State Management Tools
- **Description:** Engineered a dynamic organizational mapping solution, **increasing user productivity by 40%** and **improving workflow visualization efficiency by 50%** with React Flow and TypeScript.

Associate Software Technologist — 01/2022-05/2024

Brainwave | Human Resources

- **Role:** Backend Developer | **Tech:** Node.js, AWS (S3/Lambda/EC2), JWT, REST APIs.
- **Description:** Built a cloud-based recruitment engine, **reducing job posting time by 50%** and **improving hiring process efficiency by 35%** using AWS services and REST APIs.

mArch | Manufacturing (IoT & Logistics)

- **Role:** Backend Developer | **Tech:** Node.js, Moleculer.js (Microservices), Cube.js, Azure Services, SQL Queries.
- **Description:** Developed a high-performance logistics backend, **increasing system scalability by 45%** and **reducing service response time by 25%** using Moleculer.js microservices and SQL.

Junior Software Technologist — 06/2021-01/2022

High-Fidelity Platform Clones | E-commerce & Media

- **Role:** Full Stack Developer | **Tech:** ReactJS, Node.js, MongoDB, AWS (S3/CloudFront), Redux.
- **Description:** Developed feature-rich platform (Netflix, Twitter, Flipkart) clones, **improving frontend rendering performance by 35%** and **enhancing media streaming efficiency by 30%** using React, Redux, and AWS.

iTrac | Healthcare (Dental Informatics)

- **Role:** Lead UI Developer | **Tech:** HTML5, CSS3/SCSS, WordPress, JavaScript, Cross-Browser Compatibility.
- **Description:** Led the end-to-end development and optimization of 300+ legacy US-based dental platforms. Standardized UI components to **ensure 100% responsive design** and HIPAA-compliant accessibility.